Secure wiring and finish the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Document permanent connections and evaluate added mass or friction.

SUCCESS CRITERIA - I can...

1. Check circuit continuity and mechanical security.
2. Preserve moving-part clearance during decoration.
3. Relate a design revision to an energy transfer or force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “strain relief” and “insulation” to explain your answer. Sentence starters are optional.

strain relief: support that keeps a pull from stressing an electrical joint

insulation: material that limits unwanted electric conduction

short circuit: an unintended low-resistance path bypassing a component

01 Warm-up: bring back an earlier idea

A **40**-tooth driver turns a **240**-tooth gear. **Give ideal output**/input torque and speed ratios and say what you treated as true about energy losses.

Word help: torque = the turning effect of a force about an axis

Sentence starter: For this answer, I treated ... as true. I found ... . This supports ... because ... .

Marzano check - circle: ① ② ③

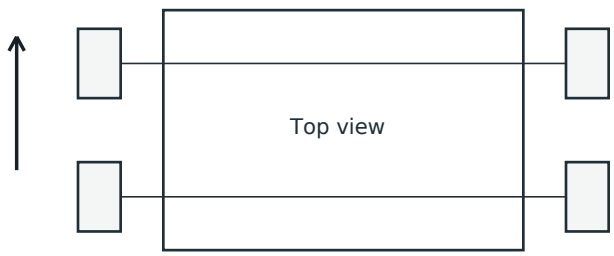
02 Worked example: explain the prediction

Reference: I = **0** with power disconnected.
Disconnect before soldering/securing wires. Use insulation and strain relief; route wires clear of gears. Follow supervised tool procedures.

Worked reasoning: A secure circuit also needs strain relief: a pull on a wire should not be carried solely by a solder joint. Insulation prevents exposed conductors from making unintended contacts.

Your turn: Draw one wire route showing a connection, insulation, strain relief and clearance from a rotating axle.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Draw your final wiring route. **Mark soldered joints**, insulation, strain relief and the switch location. **Name one possible wheel or** gear interference point.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



Marzano check - circle: ① ② ③

04 Apply the physics

Treat the car’s net forward force as a constant **0.30 N**. **Compare its acceleration** at masses of **0.20 kg** and **0.25 kg**. **Complete the table** and **calculate the percent decrease**. **Explain why this fixed-force prediction** may differ from a real motor test.

Sentence starter: Both ... . They differ in ... because ... .

Design	Mass (kg)	Net force (N)	Acceleration
Before	___	___	___
After	___	___	___

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A wire rubs a wheel. **Explain the possible effect** on energy transfer and propose a repair before another powered test.

Sentence starter: The step I would change is … . I would change it because I found … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Near an ideal long straight wire, current becomes **4** times as large and distance becomes **6** times as large.
Compare B.
Mark the distances and field directions you compare.

Word help: current = rate of charge flow

Sentence starter: Both … . They differ in … because … .

Current out of page (dot). Add field arrows.

Marzano check - circle: 1 2 3

07 Exit check

1. **Explain why** “lighter is always faster” is too strong for a powered car.
2. **Name two other influential factors**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

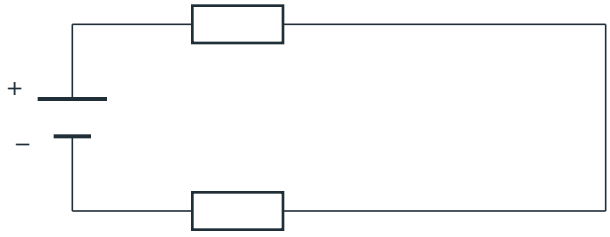


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A wire fault adds **2 ohms** in series to a motor treated as having a resistance of **4 ohms** on **6 V**.
1. **Compare current before**/after and power in the unwanted resistance.
2. **Explain why a warm joint** deserves investigation.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: 1 2 3

Read down the left column, then down the right. Use the open space to draw, calculate and explain.